helps form structures of large molecules such as protein and glycogen. In human adults total body water accounts for about 70 per cent of the lean body mass.

- **Distribution:** In an adult male of 70 kg body weight, 70, percent of water that is 30 litres is found in intracellular fluids. Of this about 4 litres are found in bones. The remaining 30 percent of water is extra cellular fluid found in 3 litres of plasma and 8.5 litres of interstitial fluid and one litre of transcellular fluid which includes saliva, pancreatic juice, aqueous humor and cerebrospinal fluid.
- **Functions:** Water performs the following functions in the body
  - Controls body temperature
  - Transports nutrients and waste
  - Dissolves important substances in tissues and cells
  - Dissolves medications
  - Lubricate the cushion joints
  - Protect the spinal cord and other sensitive tissues
  - Get rid of wastes through urination, perspiration, and bowel movements.
- **Requirements:** the requirement of water depends on person's age, weight and life style. Adults should consume 1 litre of water for every 1000kcal in their diet, infants should consume 1.5 litre/1000kcal.



Activity 3

How many glasses of water you take once a day on week days and week-ends? Check and find the difference

- **Overall water balance:** The average adult processes 2.5 litres to 3 litres of water each day.

  Water enters the body in three forms:
  - Water taken in as water or in other beverages
  - Preformed water in food
  - Metabolic water produced by cell oxidation

  Water leaves the body in the following ways via the
  - Kidneys
  - Skin
  - Lungs
  - Faeces

Water imbalance: Water can be depleted in the body due to reduced intake caused by unavailability of water, inability to obtain water and swallow it. Increased losses of water are also experienced due to hot environment, hyperventilation, prolonged vomiting and diarrhea, kidney disorders and diabetes insipidus.

- **Water deprivation:** Dehydration of body occurs when water is not taken in adequate amounts to make up for the water loss. It occurs in severe diarrhea and vomiting. Evidence of dehydration manifests in the form of sunken eyes, dry tongue, loose and inelastic skin. Simple water deprivation also causes loss of sodium and potassium. The subject should be given water, glucose and electrolytes to replenish his fluid levels.
- **Water Intoxication (water excess):** Over hydration occurs when large quantities of water are drunk in a hot climate or water excretion by the kidneys is impaired. Water Excess causes drowsiness, giddiness

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